



جامعة مصر للمعلوماتية
EGYPT UNIVERSITY
OF INFORMATICS

Egypt University of Informatics

Computer and Information Systems

Data Analysis Course

The Analysis of how University Influences Students' Access to Free Time

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Introduction

It is a well-known fact that a student's only job is to increase and improve their knowledge, whether that be in school, college, or in a job as an intern. However, no one questions at what cost does gaining knowledge stop being beneficial and start becoming detrimental to a student's overall well-being. When every waking hour is consumed by lectures, assignments and exams, the very thing that makes learning sustainable, freetime, is often the first to be sacrificed. Sleep, hobbies, social connections, and even basic rest become luxuries rather than necessities. But where exactly is the tipping point? And more importantly, does every student reach it equally?

In this survey we aim to discuss and test how university actually affects free time for students depending on their gender, major and year. Moreover, we aim to identify whether the pressure is equally distributed among students or if certain groups bear the weight more than others?

Research Question

- ★ Does university affect a student's access to free time ?and does it differ from gender, major, year?

Hypothesis

- ★ (H₀): University attendance has no effect on students' access to free time, and no significant differences exist based on gender, major, or year of study.
- ★ (H₁): University attendance reduces students' access to free time, and the reduction varies significantly by gender, major, and academic year.

Population of Interest:

- ★ The population of interest in this study is university students across Cairo, Egypt.

Sampling Method:

- ★ This population was narrowed down to university students of EUI. The data was collected using a google form which helped identify the changes in students' behaviour before and after entering university. The sampling method used is convenience sampling as the form was mainly sent to colleagues, classmates and friends.

Bias Identification:

- ★ When analysing the data we realised that 80.39% of our surveyors were 2nd year students. This was a direct cause of using convenience sampling for the survey but due to the nature of our project it was the only way we could efficiently collect enough data.
- ★ Another potential bias would be that 66.67% of our surveyors were computer science students which could create a bias towards only that major.

Survey Questions:

- ★ Gender
- ★ Major
- ★ Year
- ★ Do you have something that already takes up most of your available free time ?(work, health condition etc.)
- ★ Did you used to go out/have free time before university started?
- ★ How much free time did you have before starting university?(on avg.)
- ★ How did that change after joining university?
- ★ What (if anything) did you have to give up when university started because of the lack of time?
- ★ How long do you spend studying daily?
- ★ Do you feel that your major compared to other majors impacts your access to free time ?
- ★ How long is your commute to university?
- ★ How long does university as a whole take up from your time?

Online survey link: <https://forms.gle/UArWALgnDZrLxWZY6>

Number of samples collected: 51

Analysis:

Change in free Time based on Gender

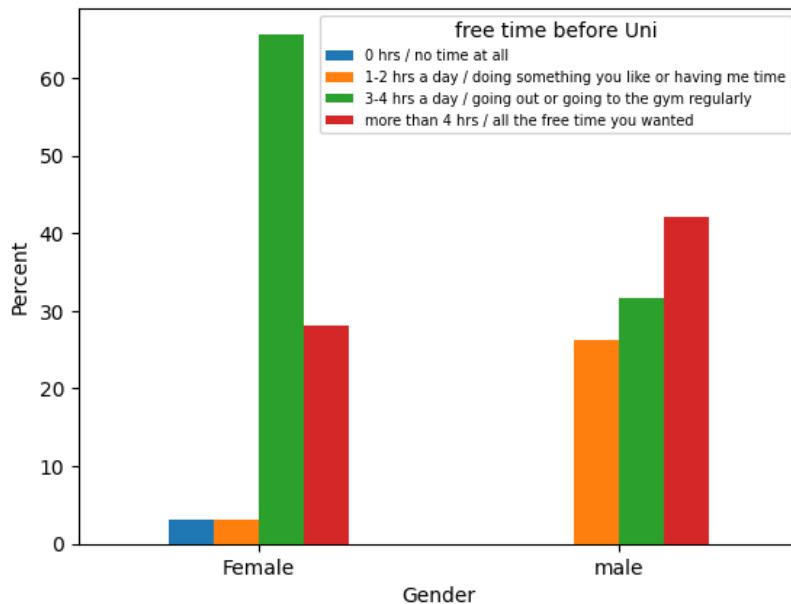
Free time before University-

Mean = 3.7 hours (1-2 hours -> 1.5 / 3-4 hours ->3.5 / 4+ hours -> 5)

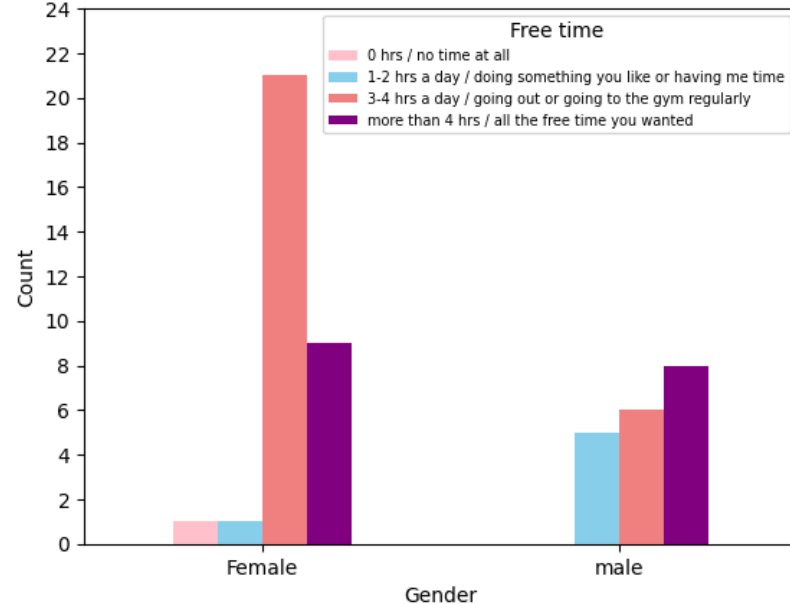
Median = 3.5 hours

Mode = '3-4 hrs a day / going out or going to the gym'

Gender vs. Free Time Before Uni

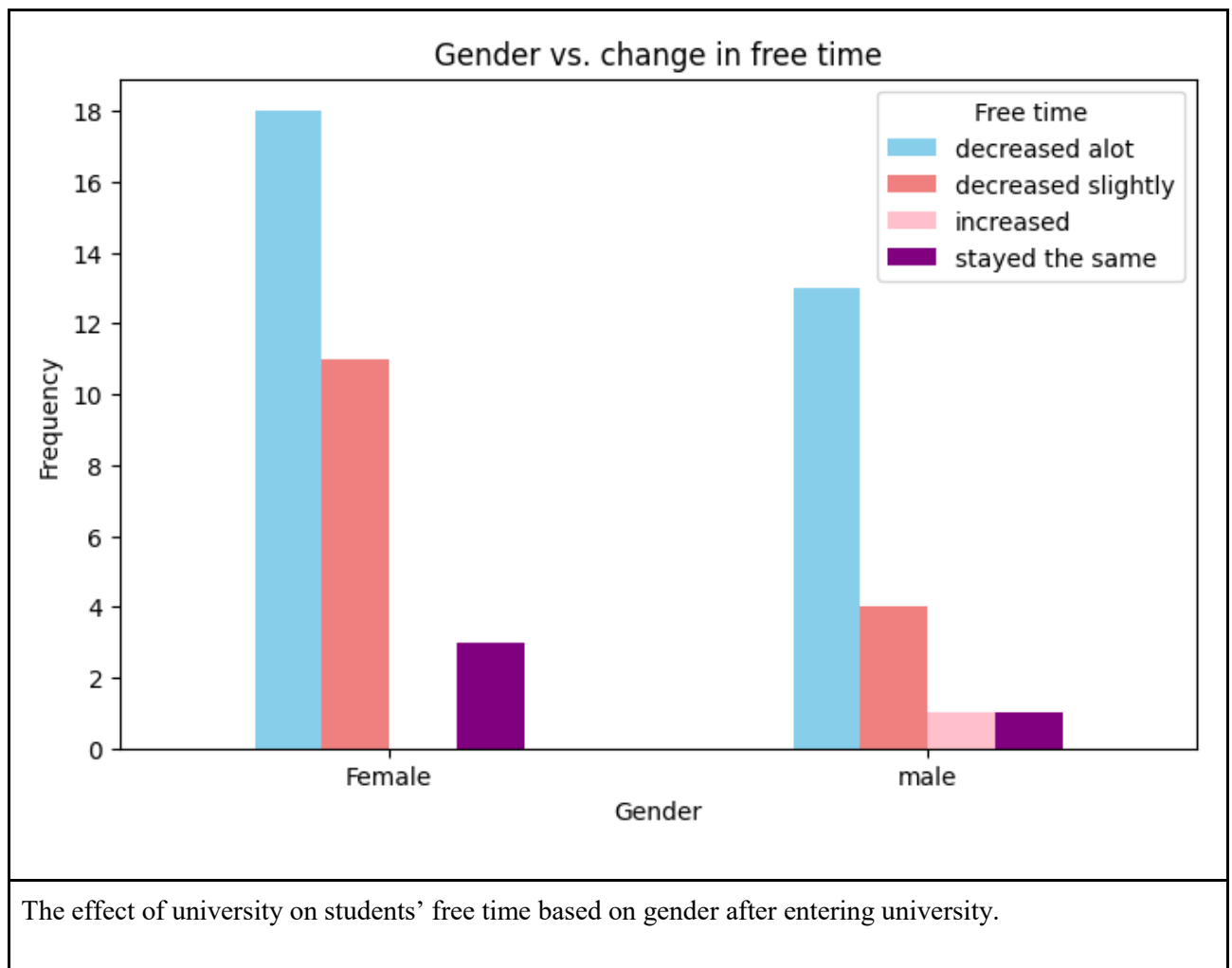


Gender vs. Free Time before Uni



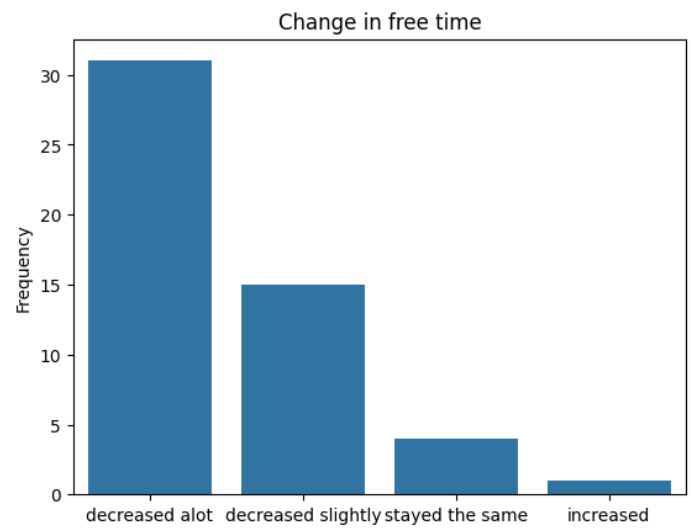
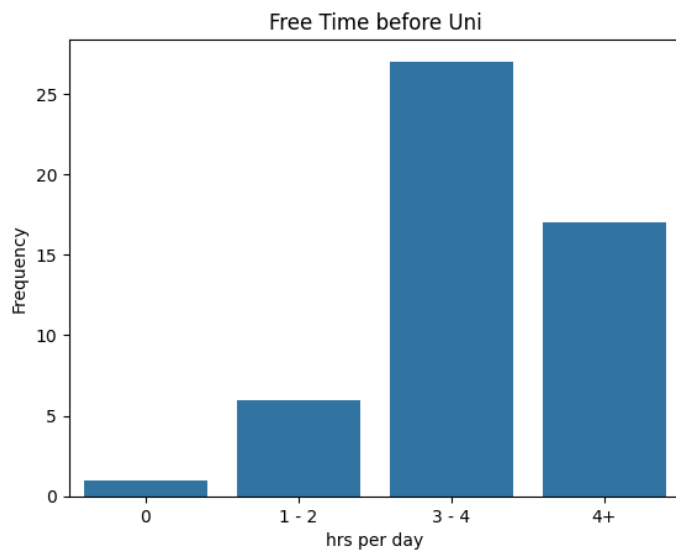
Bar Chart Showing Students free time before joining university based on gender. The first chart shows the percentage of women/men that chose each category. The second chart shows the exact frequency each option was chosen

The sample size is 51 participants with 32 women and 19 men. The questionnaire revealed that 47 people had free time before university in which they went out or pursued their hobbies, compared to only 4 who did not have much free time. **96% of women** and **86% of men** had a lot of free time. When asked how much free time they had daily, 44 people fell under the last two categories having **3+ hours** to themselves daily, which makes up **52.9%** of the sample. **65% of women** reported having had **more than 3 - 4 hours** of time and **28%** turned out to have had **more than 4 hours** daily. Similarly, **41% of men** had **3 - 4 hours daily**, while **42%** had **more than 4 hours** of personally occupied time.



56% of women reported a **strong decrease** in free time after entering university compared to **68% of men**. **34% of women** and **21% of men** reported a **slight decrease**. This shows that the majority of the sample experienced a reduction in access to free time with men experiencing a stronger effect than women.

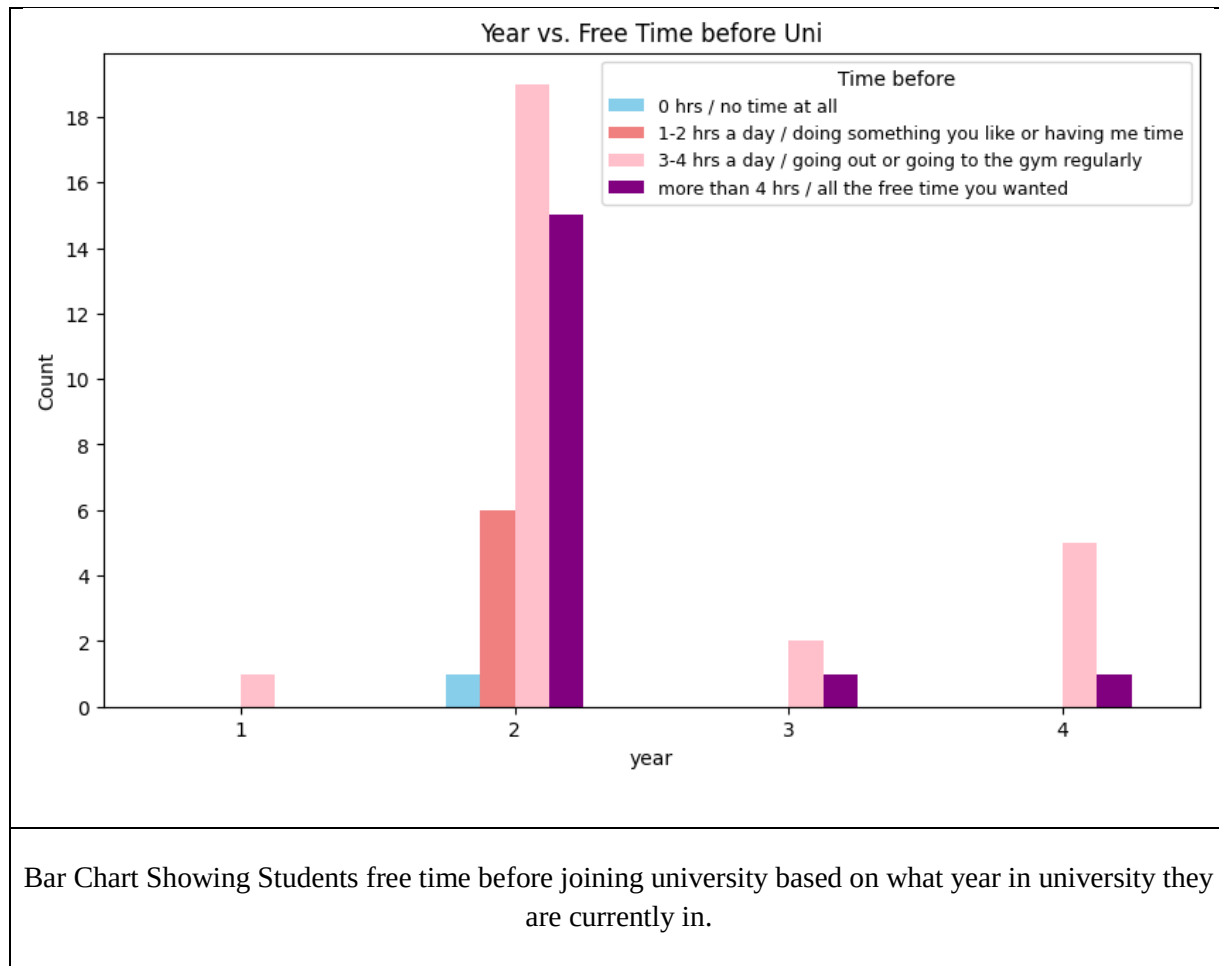
Whether this can be related to university can be examined by looking at whether the people have other priorities that occupy their time. **Almost 80% of men** answered that they have something else that takes up most of their time compared to only **56% of women**. This can be linked to the assumption that more students work to gain experience while they are still in university. Thus, it is clear that university among other external factors reduce students' access to personal and free time.



The first graph shows the available free time before uni across both genders. The second graph shows the reduction in free time across both genders.

Overall, a reduction in access to free time after entering university can be seen, regardless of gender. In summary, both genders experienced a cut back in free time after entering university with male participants having a bigger drop.

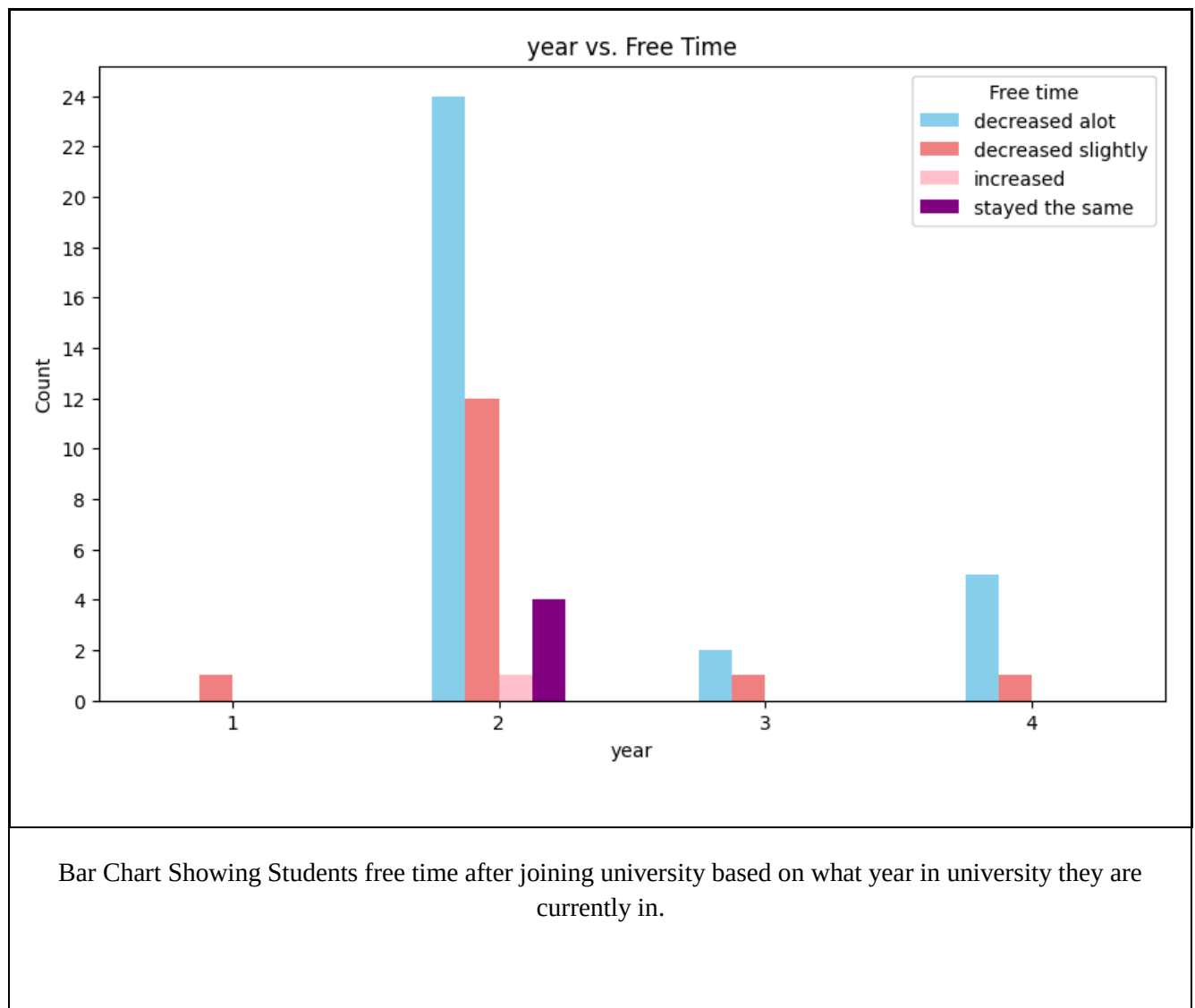
Change in free time based on academic year:



Out of the **51** respondents, the grand majority were in their second year in university coming out to be **41** students, following that only **6** were in their fourth year and the great minority were **3** students in their third year and only **1** student in their first.

When asked about their free time **before** the start of university, the **first year** student responded that they had an average **3-4** hours of free time a day. Only **2.4%** of **second year** students had **no free time**, **14.6%** had **1-2** hours, **36.6%** had **4+** hours of free time, and the majority of **46.3%** had **3-4** hours of free time, this shows that **97.6%** of **second year** students had free time before university. For the third year students, **66.7%** had **3-4** hours of free time and the rest had **4+** hours. And lastly **83.3%** of **fourth year** students had **3-4** hours and the remaining **16.7%** had **4+** hours of free time.

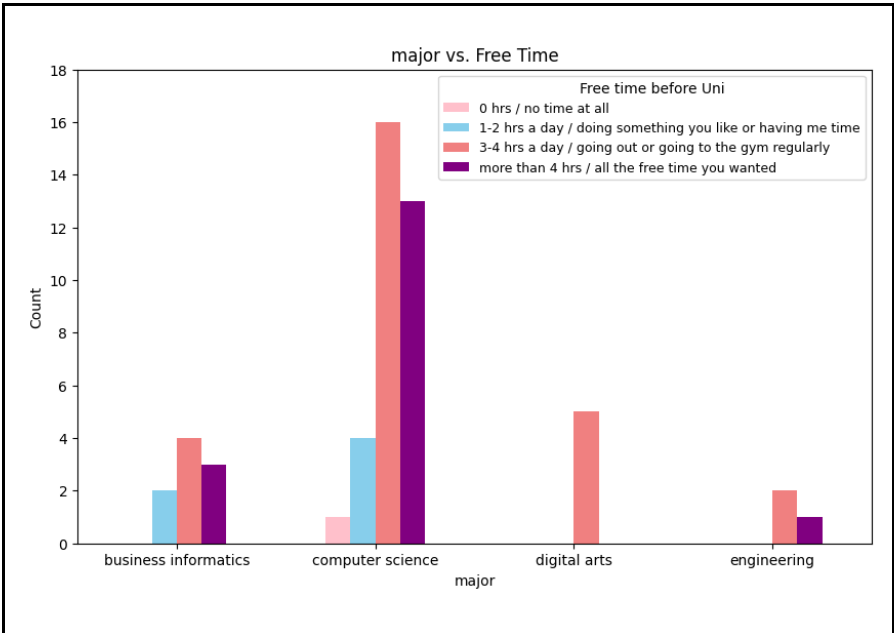
This shows that **86.3%** of students had **3+** hours of free time before entering university.



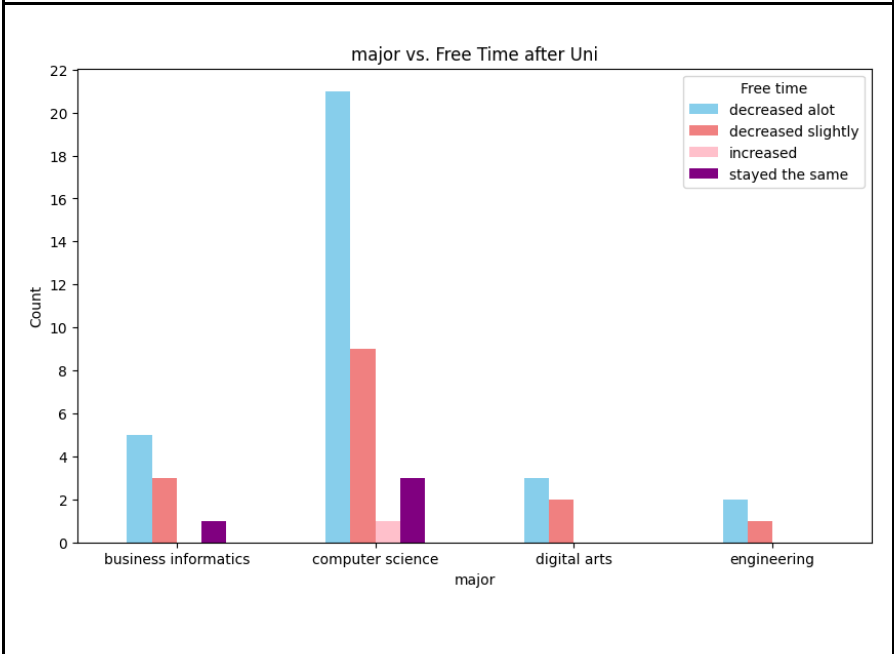
After entering university, the **first year** student mentioned a **slight decrease** in the 3-4 hours of free time. While **2.4%** of **second years** noticed an increase in free time and another minority of **9.8%** found that their free time **remained the same**, a large percentage of second year students found that their free time has **decreased** (**29.3%** only a **slight decrease** and the rest **58.5%** a **great decrease**). For the **third year** students, **33.3%** noticed a **slight decrease** in their free time and the other **66.7%** noticed a **great decrease** in their free time. **16.7%** of **fourth year** students had their free time **decrease slightly**, while the greater percentage of **83.3%** have found that their free time has **decreased greatly**.

This shows that across all academic years, a great majority of students face a decrease in their free time with **fourth year** students facing the largest change from **100%** of them having 3+ hours of free time before university to **80%+** having a **great decrease** in their free time.

Change in Free Time Based on Major:



Bar Chart Showing Students' free time before joining university across different majors.

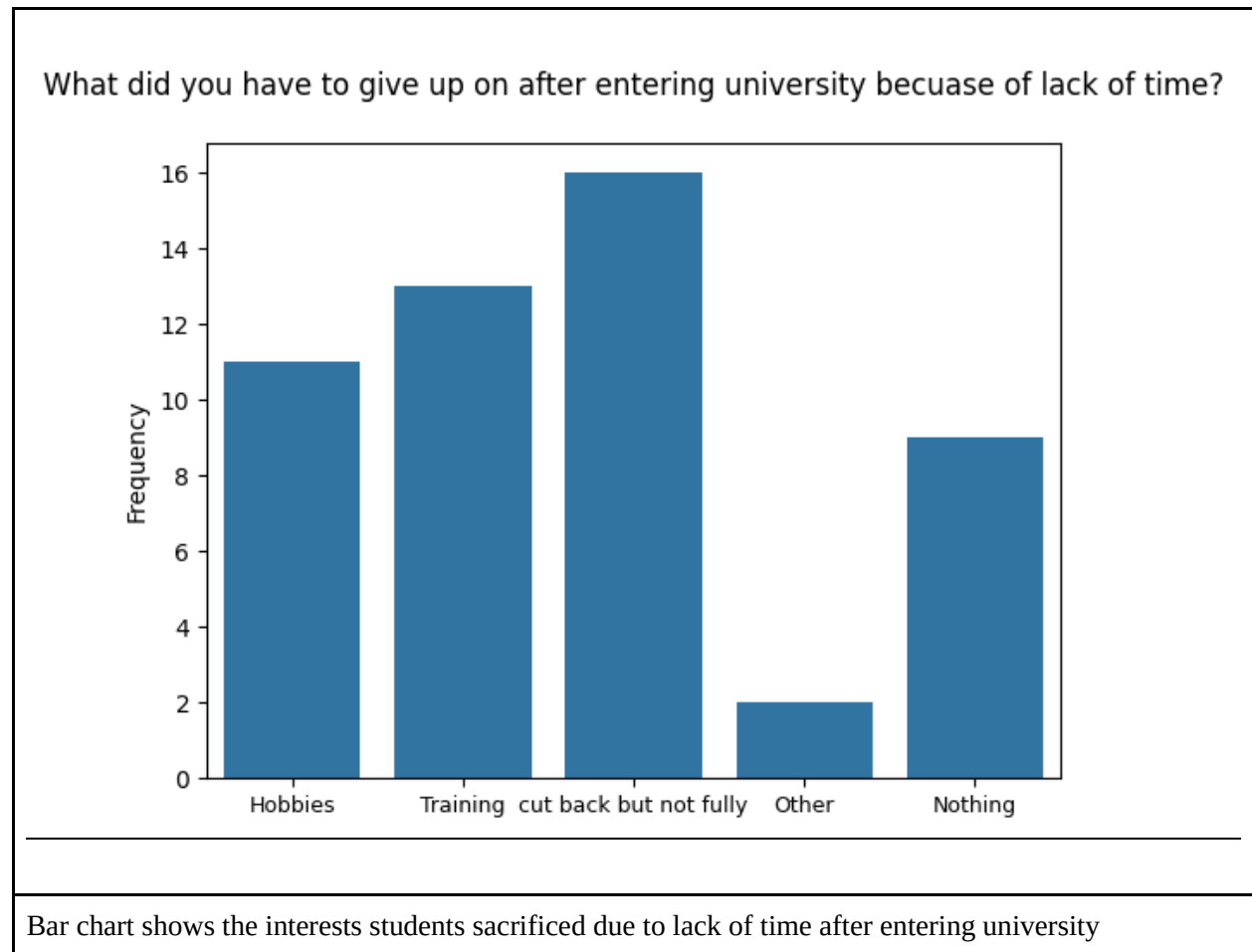


Bar Chart Showing Students' free time after joining university across different majors.

Entering university usually leads to a decrease in free time availability with approximately **92%** of all surveyed students reporting a **decrease** in their free time after entering university. Before university **84% of students** had **at least three hours** of free time daily. Computer Science students had the most relaxed initially, with **85%** having **three or more** hours of free time, but when they entered university **88%** reported that that time **decreased a lot**. Digital arts and Engineering students had a huge shift where **100% of the students** reported losing all their free time.

Engineering had the most change in free time with **66.7%** of the students reporting that their free time **decreased a lot**. Computer science follows with a percentage of **61.8%** having the most impact by amount due to the large sample size. While business and digital arts had a decrease of **55.6%** and **60%**. This shows that starting university leads to a big drop in free time for almost every student. While those in Engineering and Computer Science see the biggest change, the heavy workload across all majors makes finding personal time a constant struggle

Change in students' behavior due to lack of time:



When asked whether students had to give up anything due to lack of time after entering university, only **17%** of participants admitted to not giving up any of their personal activities. **51%** of responders had to **fully give up** on their personal activities. **31%** had to **cut back** on some activities, yet they did not give them up entirely.

This shows that due to a lack of time, which can be linked to the academic load university has on students, the vast majority of them had to sacrifice personal time and activities.

Conclusion

The findings clearly indicate that entering university causes a substantial decline in students' free time, regardless of gender, academic year or major. While the vast majority of students [86.3%] enjoyed three or more hours of free time daily before university, the time has dropped dramatically since enrollment, with approximately 92% of all respondents reporting a decrease. This shift was particularly apparent among fourth-year students and those in STEM fields such as Engineering and Computer Science, where the loss of free time was nearly universal. Even in majors like Business and Digital Arts, over half of students experienced noticeable reductions in free time. Over half the students sacrificed their personal activities after entering university. Therefore, the data proves our second hypothesis where the transition to university study, with its accompanying workload and responsibilities, leaves most students struggling to maintain personal time.

These results highlight a need for institutions to consider strategies that better support student well-being and time management, as the current academic environment appears to systematically ruin a critical resource for rest and personal development.

Any potential issues

- ★ Due to our bias (overrepresentation of computer science and 2nd year students) this could lead to a misrepresentation for a major or year since we don't have enough data to sufficiently test our hypothesis unbiasedly on.